



DEFENDTHEDOMINION.COM
45°25'26.53"N 75°41'43.79"W

ArcticSim

Hack The North

DOMINION
DYNAMICS

▶ Hack the North Competition Scenario

Northern Sea Route increasingly used by Russia's shadow fleet



By The Watch – January 30, 2026 – Updated: March 12, 2026 – No Comments – 3 Mins Read



LNG Shadow Fleet on Route to the Arctic



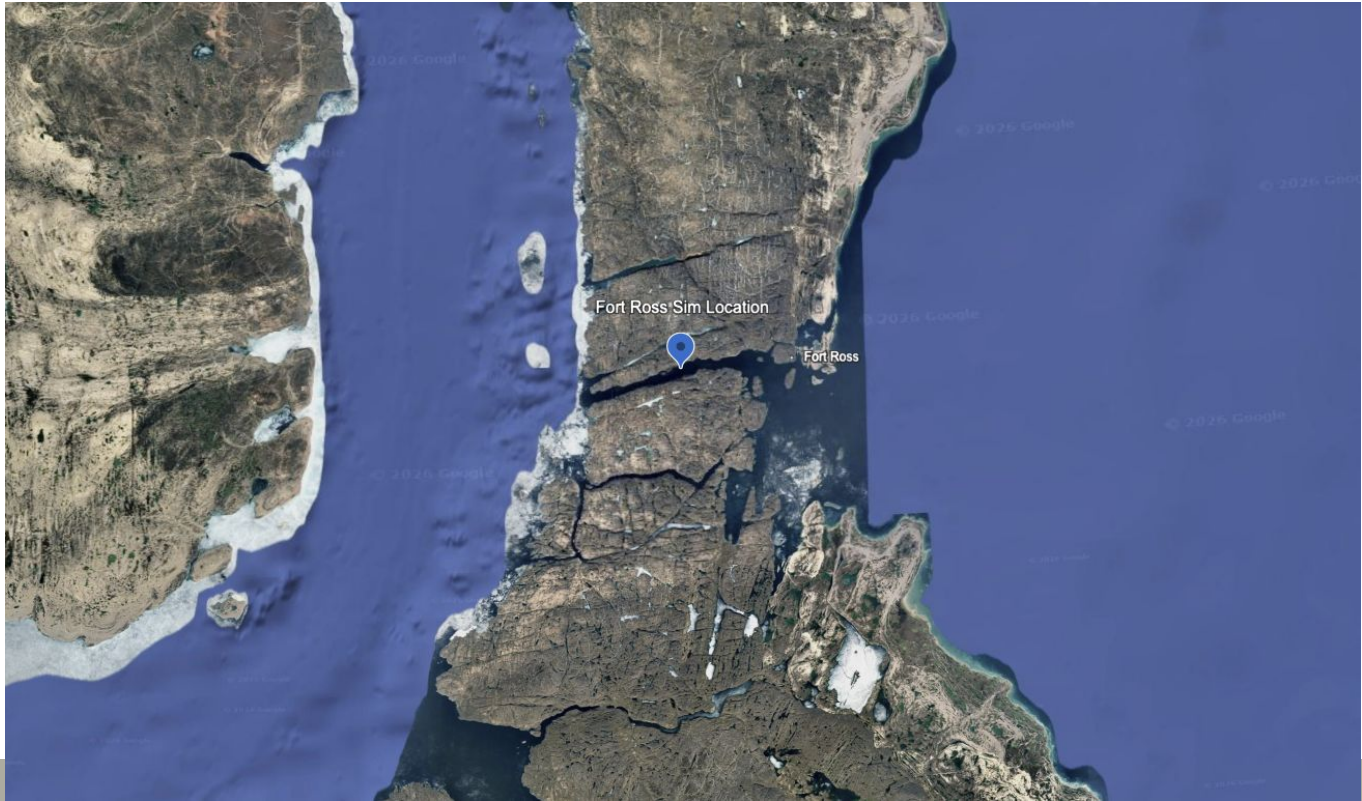
Source: Ship tracking data compiled by Bloomberg

Bloomberg

Arctic - Northwest Passage



Arctic - Bellot Strait (Near Fort Ross)



Fort Ross Bellot Strait

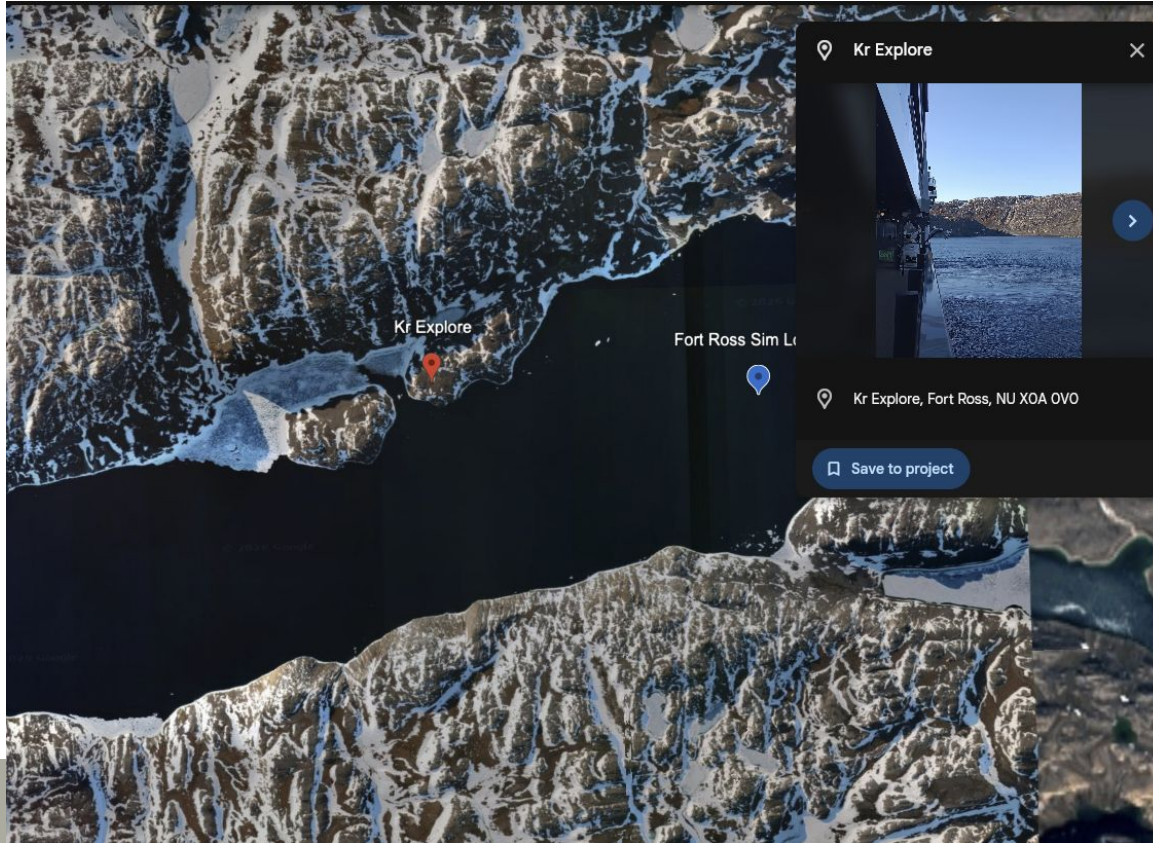


Bellot Strait, via Seemanns. 1852.

- Bellot Strait was discovered 1852
- Fort Ross (HBC trading post) built in 1937, abandoned 1948
- In 1942 the first west-end transit of NWP used this strait
- In 1943, Fort Ross was the LZ for the first parachute jump in the Arctic circle

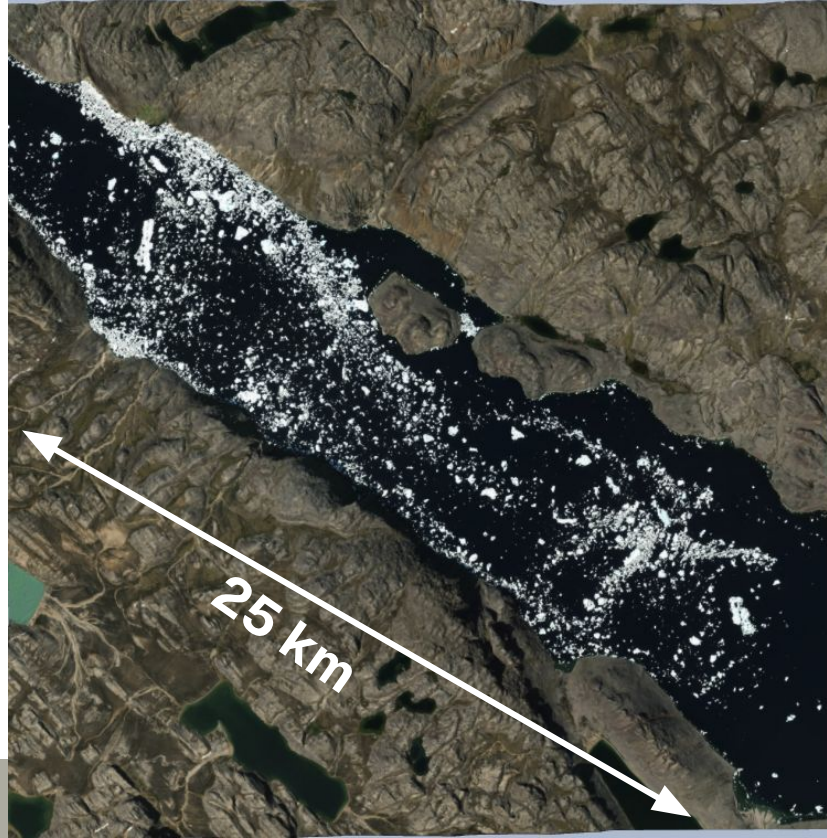


Commercial / Civ Transit



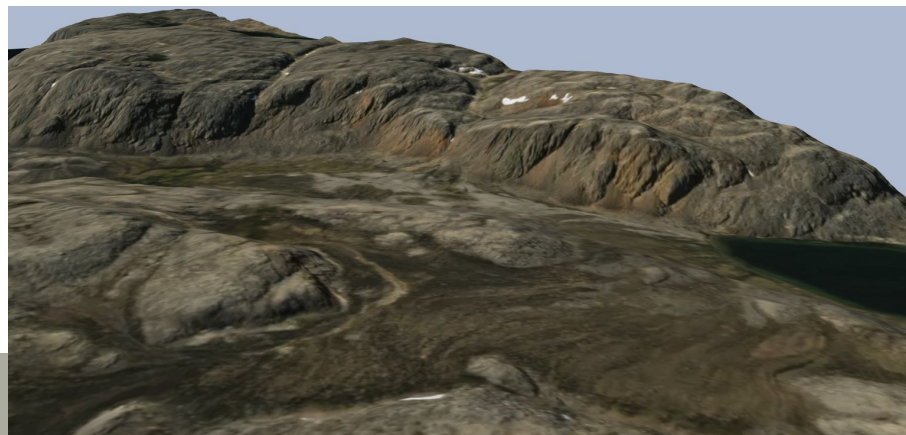
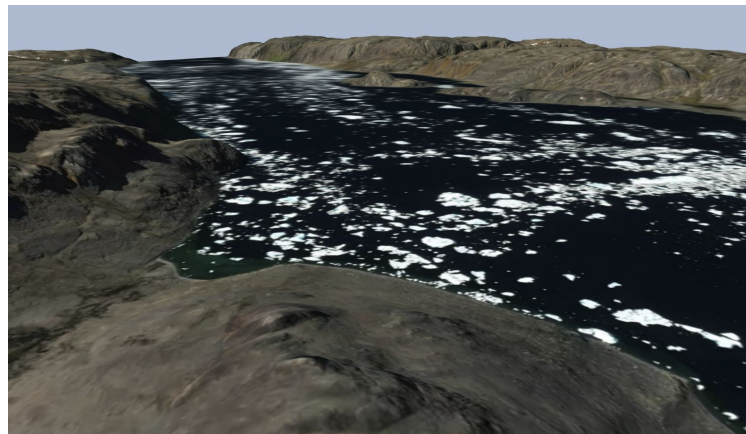
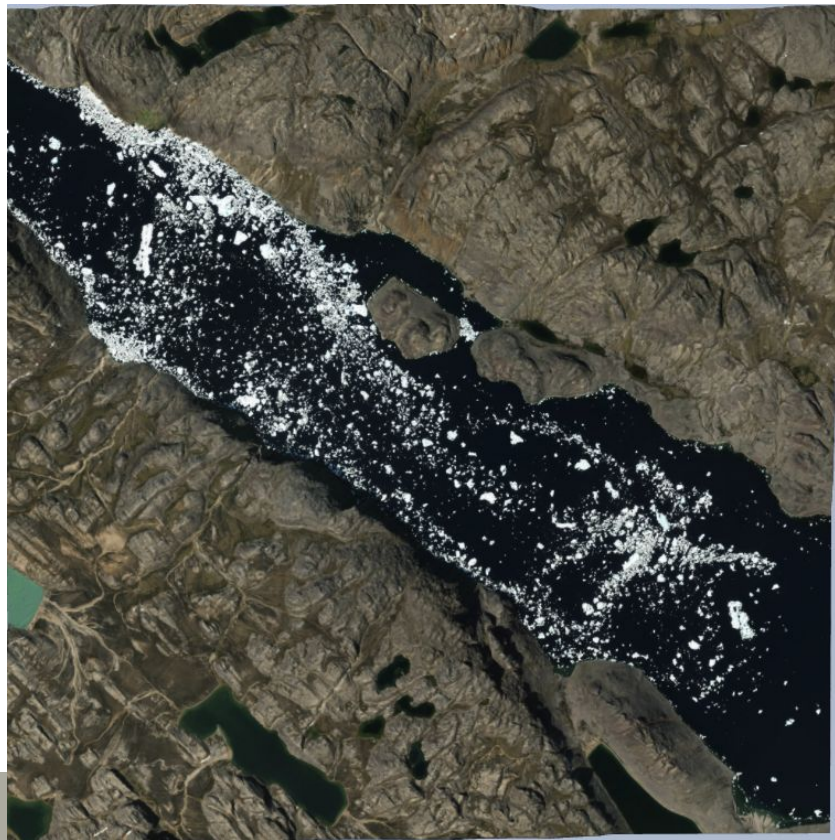
Arctic - Northwest Passage

2km



actual sim
render

Arctic Sim Terrain



► Goal: Detect & Track the movement of this shadow vessel

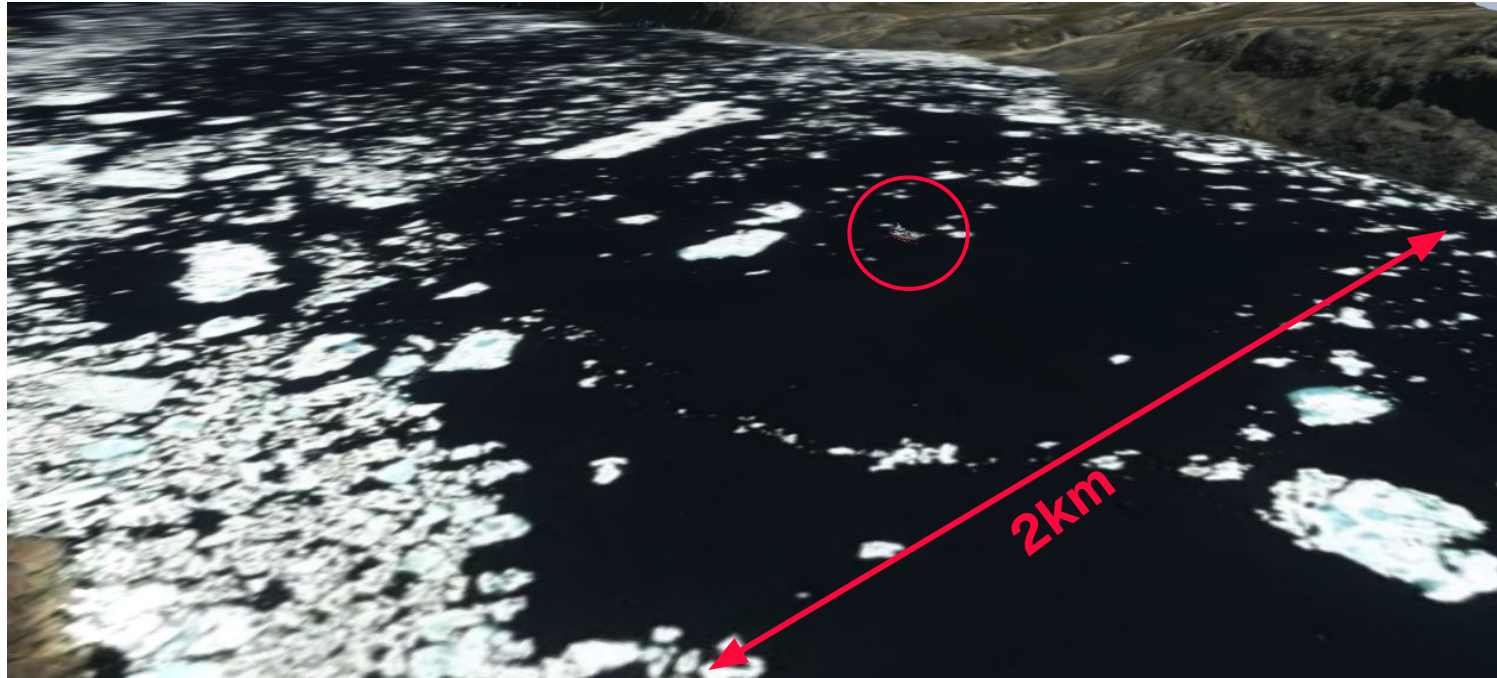
Randomly
spawns



No AIS

Random
movement
path

▶ Goal: To Scale



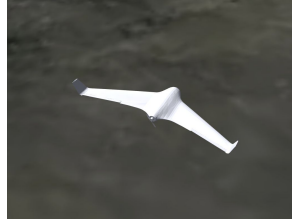
► You are given: 4 Autonomous Drone Assets

tower 1



quadcopter 1

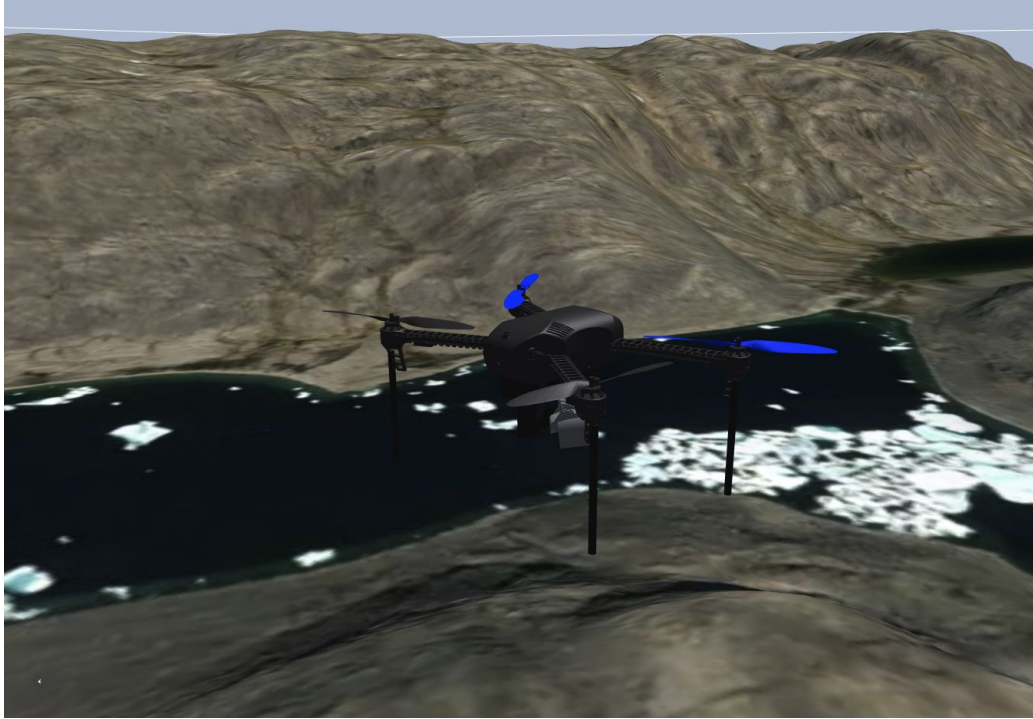
tower 2



fixed-wing 1

Teams will be judged on *search efficiency, coverage, detection speed, tracking duration, accuracy, autonomy & collaboration*

◀ QuadCopter Drone

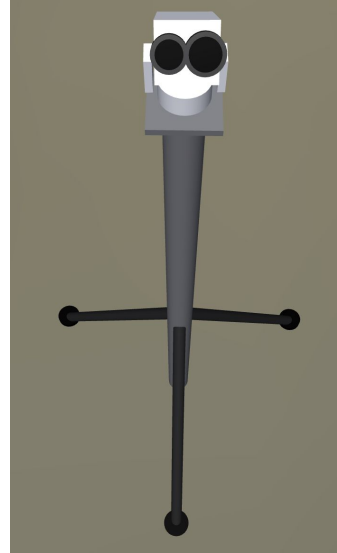


ARDUPILOT

ArduPilot Copter

 **MAVLINK**
MICRO AIR VEHICLE COMMUNICATION PROTOCOL

▶ Tower Drone

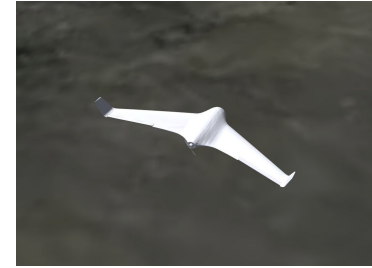
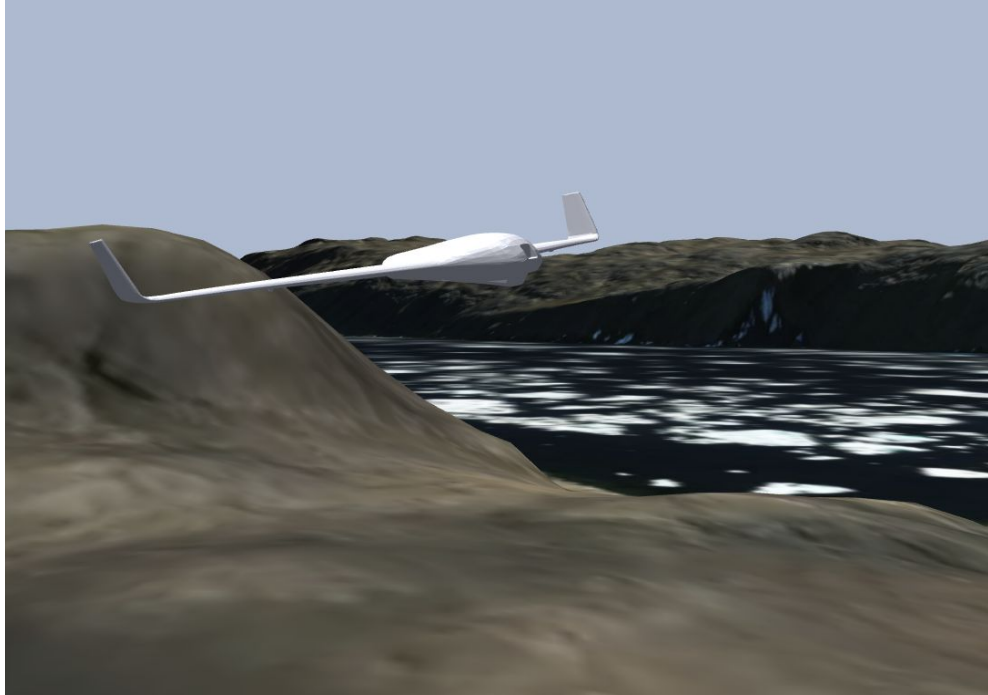


ARDUPILOT

ArduPilot AntennaTracker

 **MAVLINK**
MICRO AIR VEHICLE COMMUNICATION PROTOCOL

Fixed-Wing Drone



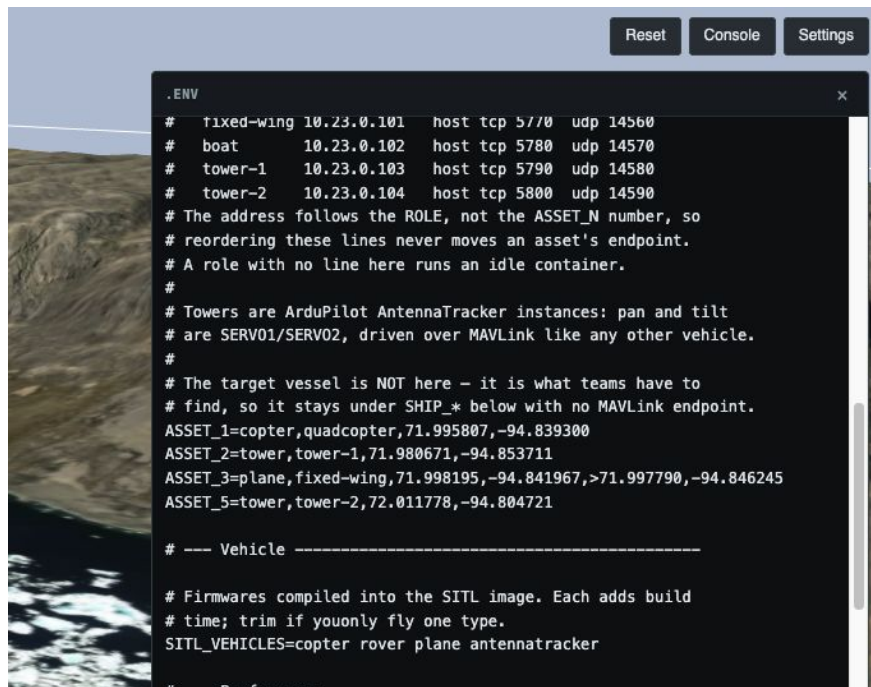
ARDUPILOT

ArduPilot Plane

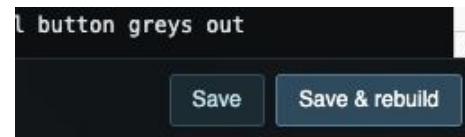
 **MAVLINK**
MICRO AIR VEHICLE COMMUNICATION PROTOCOL

◀ You can place anywhere (locations set via .env in Settings)

comment out
what you don't
need

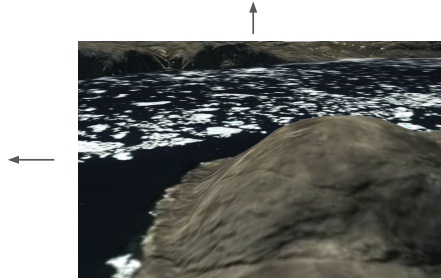


save & rebuild
will restart all
docker
containers

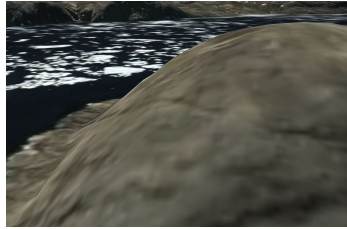


do not play with .env too much! You can break your sim with this

▶▶ How to Navigate the Sim



→ **Pan** = Left Click + Drag



Zoom = Scrolling

Rotate = CTRL Button +
Left Click Drag



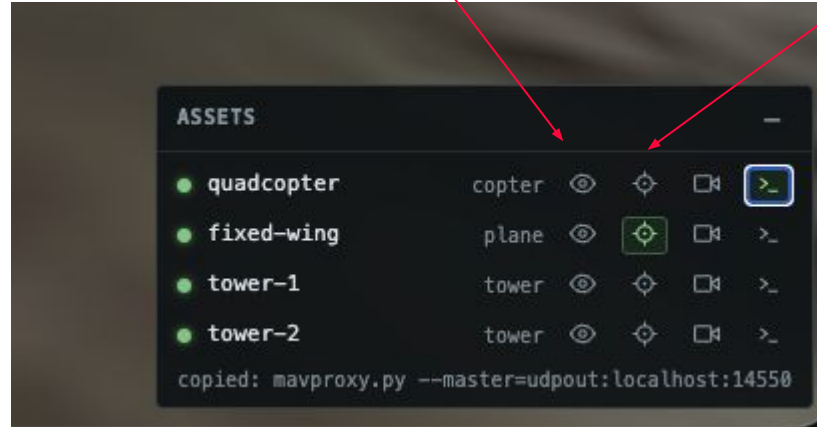
If you get lost press the “Eye ball”



▶ Fly-to & Follow Assets in 1 Click

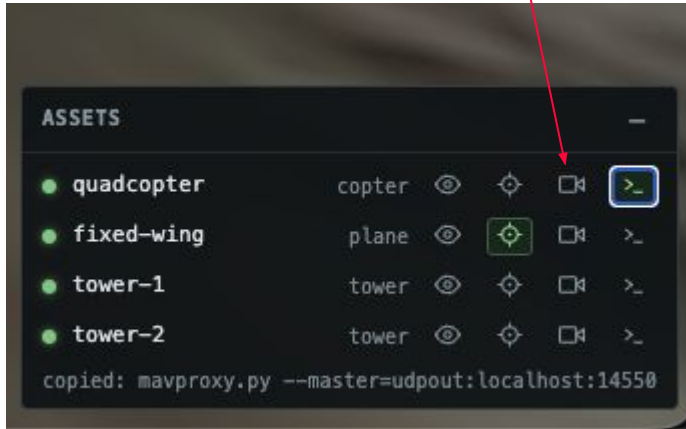
sim cam goto asset

sim cam follows asset



▶ Asset Cameras are 1 Click Away

opens asset cam

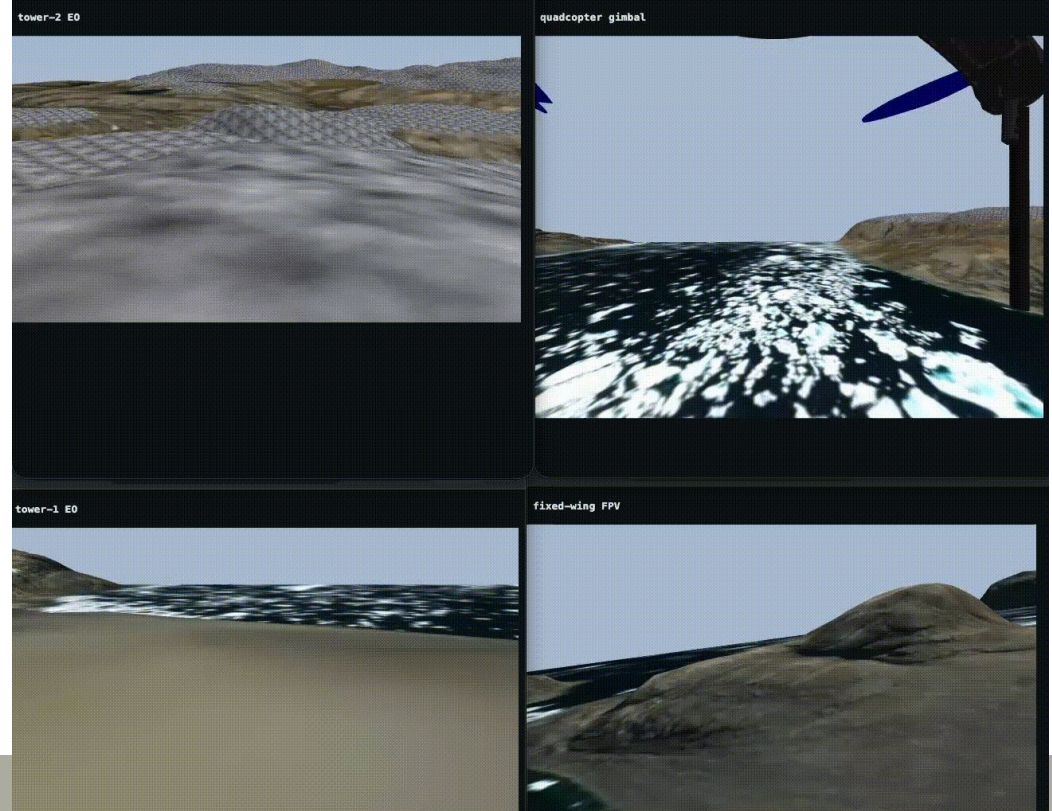


Field of Views & Resolutions

quad camera HFOV 114.6° VFOV ~99.4° @ 640x480

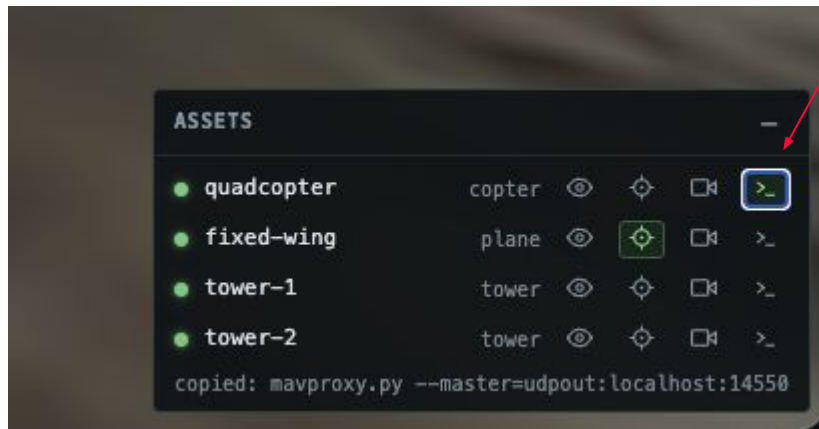
fixed-wing HFOV 69.0° VFOV ~42.6° @ 640x360

towers HFOV 60.0° VFOV ~36.1° @ 640x360



MAVProxy Console Access is 1 Click Away

- view mavlink telemetry
- issue mavlink commands



```
python -m venv .venv
source .venv/bin/activate
pip install gnureadline
pip install billiard
pip install future
pip install numpy pyparsing
pip install MAVProxy
```

```
+ ~ mavproxy.py --master=udpout:localhost:14550
Connect udpout:localhost:14550 source_system=255
Failed to load module: No module named 'adsb'. Use 'set moddebug 3' in the MAVProxy console to enable traceback
Log Directory:
Telemetry Log: mav.tlog
Waiting for heartbeat from localhost:14550
MAV: Detected vehicle 1:1 on link 0
online system 1
STABILIZE> Mode STABILIZE
fence present
AP: ArduCopter V4.5.7 (b7207551)
AP: c253b9f5a5f533d5efc99af96a7696b6
AP: Frame: QUAD/X
Received 1433 parameters (ftp)
Saved 1433 parameters to mav.parm
STABILIZE> Flight battery 100 percent

STABILIZE>
STABILIZE>
STABILIZE>
STABILIZE> mode guided
STABILIZE> Got COMMAND_ACK: DO_SET_MODE: ACCEPTED
GUIDED> Mode GUIDED

GUIDED>
GUIDED>
GUIDED> arm throttle
GUIDED> Got COMMAND_ACK: COMPONENT_ARM_DISARM: ACCEPTED
AP: Arming motors

GUIDED>
GUIDED> ARMED
takeoff 100
GUIDED> Take Off started
Got COMMAND_ACK: NAV_TAKEOFF: ACCEPTED
AP: EKf3 IMU0 MAG0 in-flight yaw alignment complete
AP: EKf3 IMU1 MAG0 in-flight yaw alignment complete
height 15
height 20
height 35
height 45
height 55
height 65
height 75
height 85
height 95
Flight battery 100 percent
Flight battery 100 percent
Flight battery 100 percent
Flight battery 100 percent
```

▶ You will crash - to restart your sim and all assets - PRESS “Reset”



◀ QuadCopter MAVProxy Basics



Takeoff in 3 commands

```
mode guided
```

```
arm throttle
```

```
takeoff 100
```

note: only runs for 3 secs
so issue the takeoff
quickly

altitude

Fly somewhere in 1 command

```
guided <lat> <lon> <alt>
```



Learn more: <https://ardupilot.org/copter/index.html>

Learn more: <https://ardupilot.org/mavproxy/>

right click in your sim
to copy coordinates

Fixed Wing MAVProxy Basics



Takeoff in 3 commands

```
mode guided
```

```
arm throttle
```

```
mode takeoff
```

note: only runs for 3 secs
so issue the takeoff
quickly

will circle around

Fly somewhere in 1 command

```
guided <lat> <lon> <alt>
```



Learn more: <https://ardupilot.org/plane/index.html>

Learn more: <https://ardupilot.org/mavproxy/>

right click in your sim
to copy coordinates

▶ Tower MAVProxy Basics



Pan + Tilt Camera

servo set 1 1900 #pan (yaw)

servo set 2 1200 #tilt (pitch)

Randomly Scan

mode scan

Learn more: <https://ardupilot.org/antennatracker/>

Learn more: <https://ardupilot.org/mavproxy/>

◀ Sending Boat Tracking Details

Confirm with DD staff
once you submit a test
tracking detection.

Create a boat tracking hit

```
curl -X POST http://<SIM-IP>:8010/api/tracks \  
-H 'content-type: application/json' \  
-d '{"name":"Sierra One","lat":71.9965,"lon":-94.8448}'
```

```
{"ok":true,"created":true,"name":"Sierra One","uuid":"entity-22ce689495","lat":71.9965,"lon":-94.8448,"...
```

Update the boats location (name must match)

```
curl -X POST http://<SIM-IP>:8010/api/tracks \  
-H 'content-type: application/json' \  
-d '{"name":"Sierra One","lat":71.9975,"lon":-94.8450,"heading":315,"speed":6.5}'
```

```
{"ok":true,"created":false,"name":"Sierra One","uuid":"entity-22ce689495","lat":71.9975,"lon":-94.845,"...
```

List tracks

```
curl http://10.99.0.1:8010/api/tracks
```



Next Steps

```
$ git clone https://github.com/Dominion-Dynamics/arctic-sim  
$ cd arctic-sim  
$ cp .env.example .env  
$ docker compose up --build
```

Deploy your arctic-sim locally if you meet system requirements

8 Cores (+12 recommended)
16GB RAM (+24 recommended)
50 GB Disk

<https://www.mapbox.com/> Get a free MAPBOX_TOKEN (without it you will have low-res terrain images)

If you do not meet system requirements, tell us who will be in your team (max 10) and we will give you a wireguard config to connect to a dedicated cloud instance.

Wireguard install <https://www.wireguard.com/install/>

Happy Hunting.